Step 1 — Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2 — Load Dataset

In [2]:
```
df = pd.read_csv("../data/DisasterDeclarations.csv")
df.head()
```

Out[2]:

	femaDeclarationString	disasterNumber	state	declarationType	declarationDate	fyDecla
0	FM-5529-OR	5529	OR	FM	2024-08-09T00:00:00.000Z	2
1	FM-5528-OR	5528	OR	FM	2024-08-06T00:00:00.000Z	2
2	FM-5527-OR	5527	OR	FM	2024-08-02T00:00:00.000Z	2
3	DR-4312-CA	4312	CA	DR	2017-05-02T00:00:00.000Z	2
4	DR-4251-AL	4251	AL	DR	2016-01-21T00:00:00.000Z	2

5 rows × 28 columns

Step 3 — Validate Time Data

In [3]:
```
df['declarationDate'] = pd.to_datetime(df['declarationDate'])
df['incidentBeginDate'] = pd.to_datetime(df['incidentBeginDate'])
df['incidentEndDate'] = pd.to_datetime(df['incidentEndDate'])
```

In [4]:
```
df['year'] = df['declarationDate'].dt.year
df['month'] = df['declarationDate'].dt.month
```

In [5]:
```
df['declarationDate'].isnull().sum()
```

Out[5]:
```
np.int64(0)
```

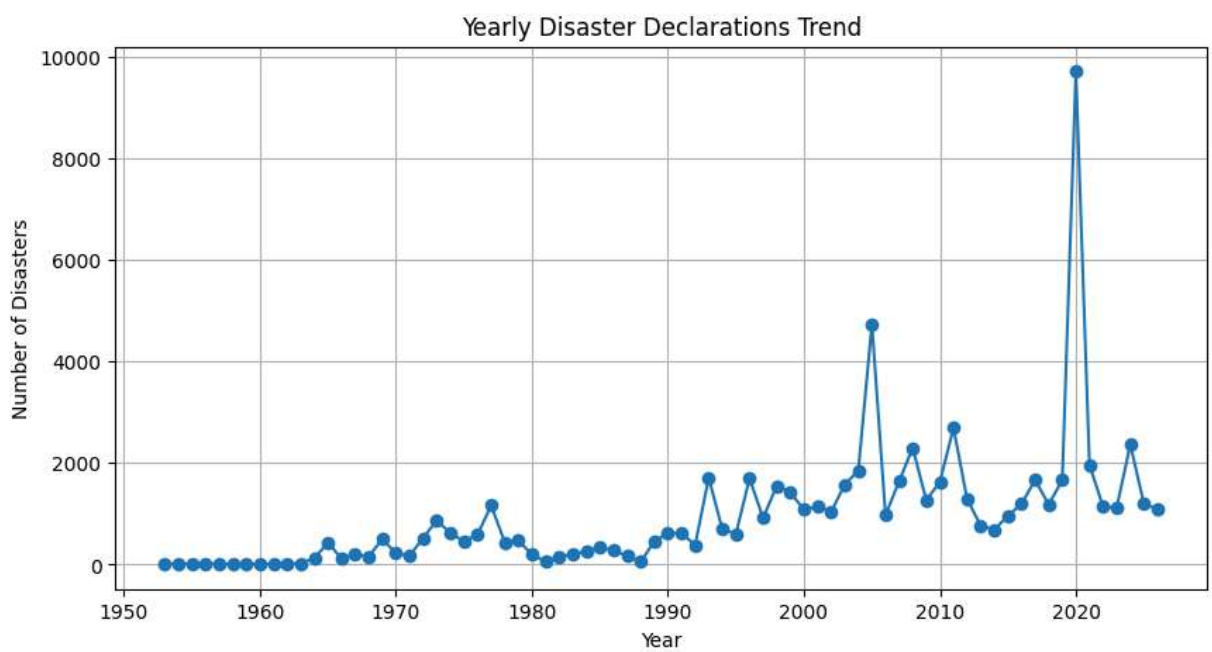
Step 4 — Yearly Trend Analysis Group disasters by year.

In [6]:
```
yearly_disasters = df.groupby('year').size()
yearly_disasters.head()
```

Out[6]:
```
year
1953    13
1954    17
1955    18
1956    16
1957    16
dtype: int64
```

Visualization

In [7]:
```
plt.figure(figsize=(10,5))

plt.plot(yearly_disasters, marker='o')

plt.title("Yearly Disaster Declarations Trend")
plt.xlabel("Year")
plt.ylabel("Number of Disasters")

plt.grid(True)

plt.show()
```



Observation: The plot shows how disaster declarations change over time.

Pattern: There is an increasing trend in disaster declarations in recent years.

Possible Reason: Climate change and improved disaster reporting systems.

Step 5 — Incident Type Trends Over Time

Group by year and incident type

In [8]:
```
incident_trend = df.groupby(['year','incidentType']).size().unstack()
incident_trend.head()
```

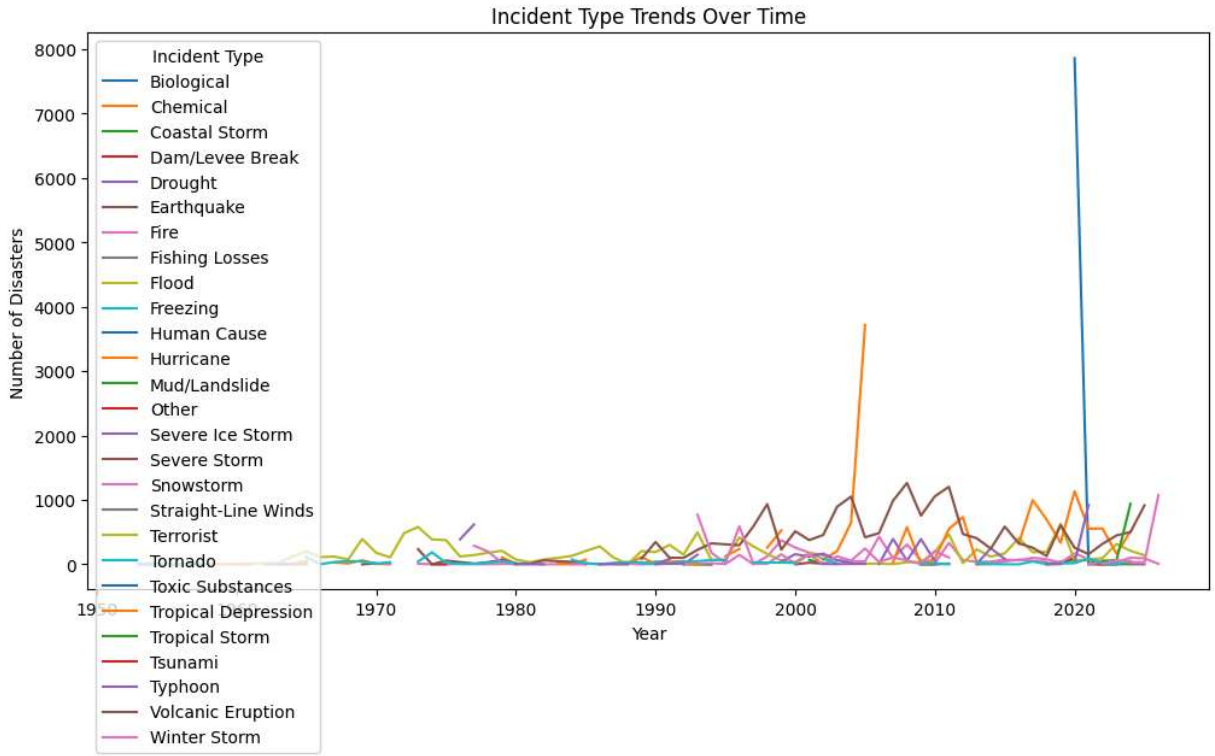
Out[8]:

incidentType	Biological	Chemical	Coastal Storm	Dam/Levee Break	Drought	Earthquake	Fire	Fishir Loss
year								
1953	NaN	NaN	NaN	NaN	NaN	NaN	1.0	Na
1954	NaN	NaN	NaN	NaN	NaN	1.0	NaN	Na
1955	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Na
1956	NaN	NaN	NaN	NaN	NaN	NaN	1.0	Na
1957	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Na

5 rows × 27 columns

Visualization

In [9]:
```
incident_trend.plot(figsize=(12,6))

plt.title("Incident Type Trends Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Disasters")

plt.legend(title="Incident Type")

plt.show()
```



Observation: Different disaster types vary across years.

Pattern: Storms and floods appear more frequently than other disaster types.

Possible Reason: Extreme weather events are becoming more common.

Step 6 — Monthly Seasonality Analysis

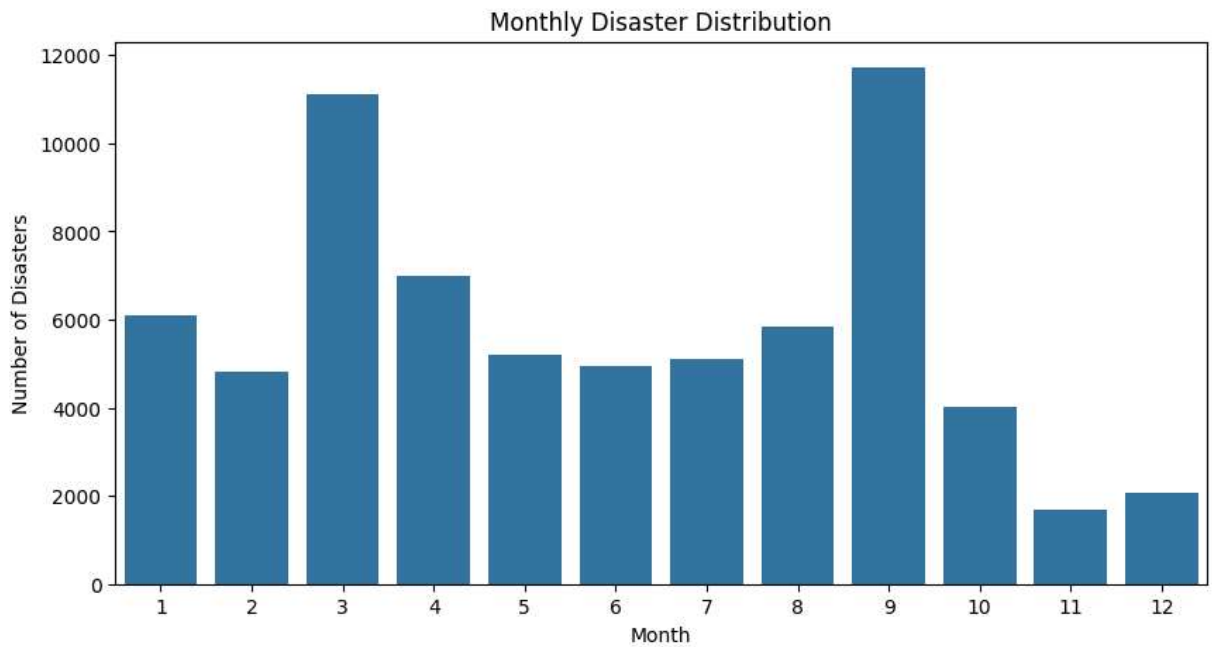
Group by month.

In [10]:
```
monthly_disasters = df.groupby('month').size()
monthly_disasters
```

Out[10]:
```
month
1      6111
2      4830
3     11127
4      6998
5      5211
6      4931
7      5101
8      5849
9     11721
10     4007
11     1674
12     2074
dtype: int64
```

visualization

In [11]:
```
plt.figure(figsize=(10,5))

sns.barplot(x=monthly_disasters.index, y=monthly_disasters.values)

plt.title("Monthly Disaster Distribution")
plt.xlabel("Month")
plt.ylabel("Number of Disasters")

plt.show()
```



insight

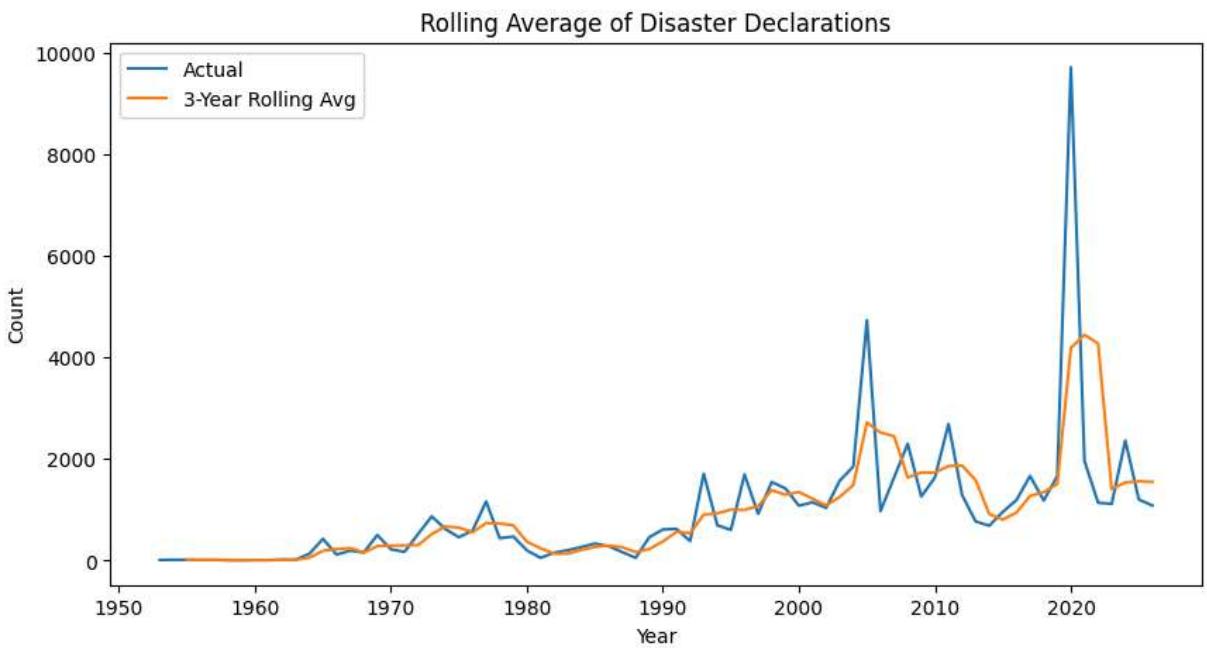
Observation: Some months show higher disaster declarations.

Pattern: Summer and hurricane season months tend to have more disasters.

Possible Reason: Seasonal weather patterns like hurricanes and floods.

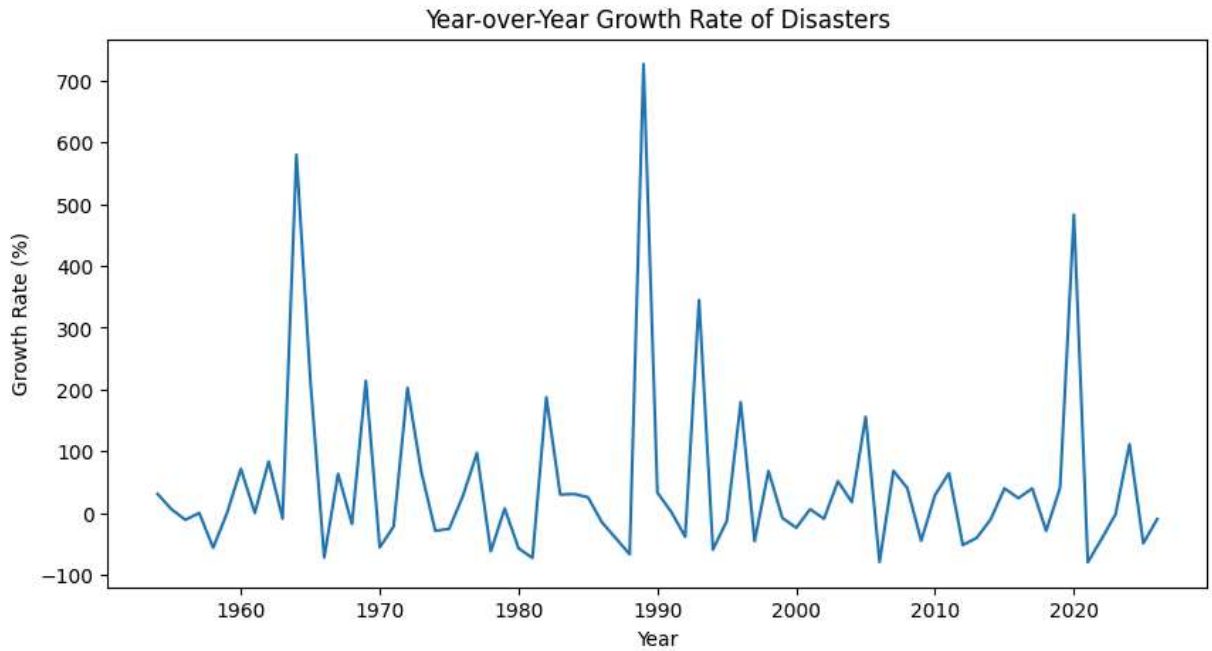
Step 7 — Advanced Time Insights Rolling Average (3-year smoothing)

In [12]:
```
rolling_avg = yearly_disasters.rolling(window=3).mean()

plt.figure(figsize=(10,5))

plt.plot(yearly_disasters, label="Actual")
plt.plot(rolling_avg, label="3-Year Rolling Avg")

plt.title("Rolling Average of Disaster Declarations")
plt.xlabel("Year")
plt.ylabel("Count")

plt.legend()

plt.show()
```



Growth Rate (Year-over-Year)

In [13]:
```
growth_rate = yearly_disasters.pct_change()*100

plt.figure(figsize=(10,5))

plt.plot(growth_rate)

plt.title("Year-over-Year Growth Rate of Disasters")
plt.xlabel("Year")
plt.ylabel("Growth Rate (%)")

plt.show()
```



Step 8 — Identify Peak Disaster Years

In [14]:
```
top_years = yearly_disasters.sort_values(ascending=False).head(5)

top_years
```

Out[14]:
```
year
2020    9712
2005    4736
2011    2690
2024    2364
2008    2300
dtype: int64
```

Step 9 — Final Insight Summary

Key Findings:

1. Disaster declarations have increased over the past decades.
2. Storms and floods are the most common disaster types.
3. Certain months show clear seasonal patterns.
4. Rolling averages confirm a long-term upward trend.

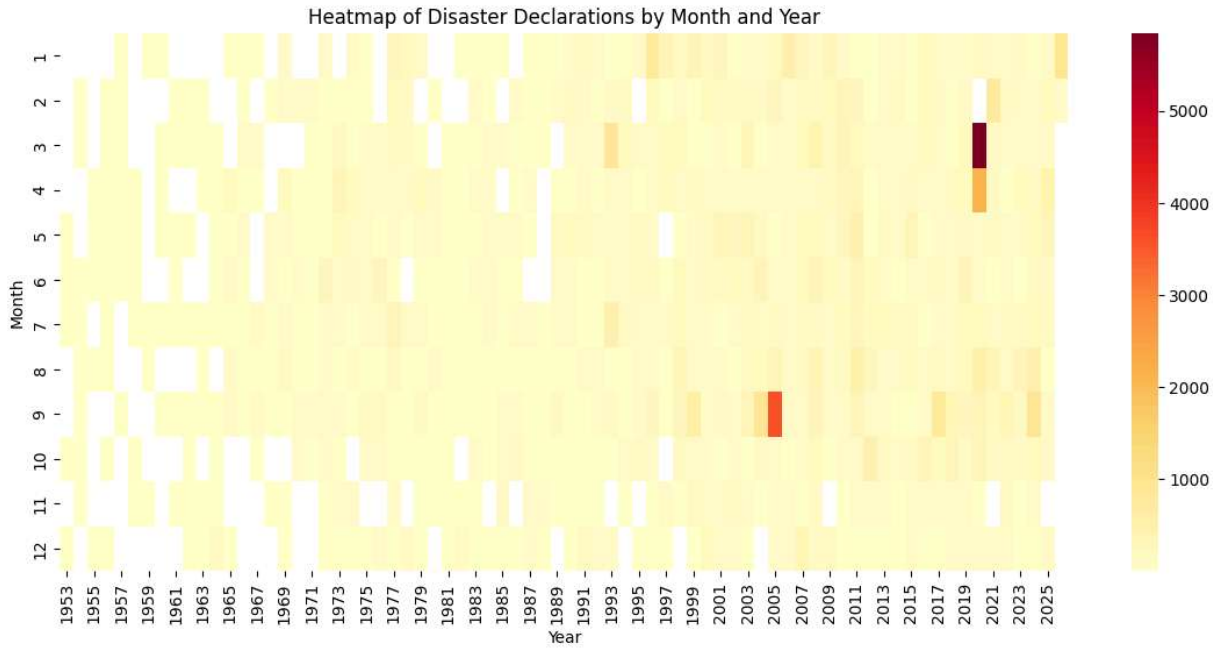
Heatmap of Disasters by Year and Month

Step 1 — Create Pivot Table

In [15]:
```
pivot_table = df.pivot_table(
    index='month',
    columns='year',
    aggfunc='size'
)
```

Step 2 — Create Heatmap

In [16]:
```
plt.figure(figsize=(14,6))

sns.heatmap(pivot_table, cmap="YlOrRd")

plt.title("Heatmap of Disaster Declarations by Month and Year")
plt.xlabel("Year")
plt.ylabel("Month")

plt.show()
```



What This Graph Shows

-->Dark colors → More disasters

-->Light colors → Fewer disasters

You can easily see:

*seasonal patterns

*years with many disasters

*months with highest disaster frequency

Observation: The heatmap visualizes disaster declarations across months and years.

Pattern: Certain months consistently show higher disaster counts.

Interpretation: Seasonal weather events such as hurricanes and storms contribute to higher disaster frequency in specific months.

Top 10 States Disaster Trend Over Time (Multi-Line Chart).

Step 1 — Find Top 10 States with Most Disasters

In [17]:
```
top_states = df['state'].value_counts().head(10).index

top_states
```

Out[17]:
```
Index(['TX', 'KY', 'MO', 'FL', 'GA', 'VA', 'LA', 'OK', 'NC', 'PR'], dtype='objec
t', name='state')
```

Step 2 — Filter Dataset for These States

In [18]:
```
top_states_df = df[df['state'].isin(top_states)]
```

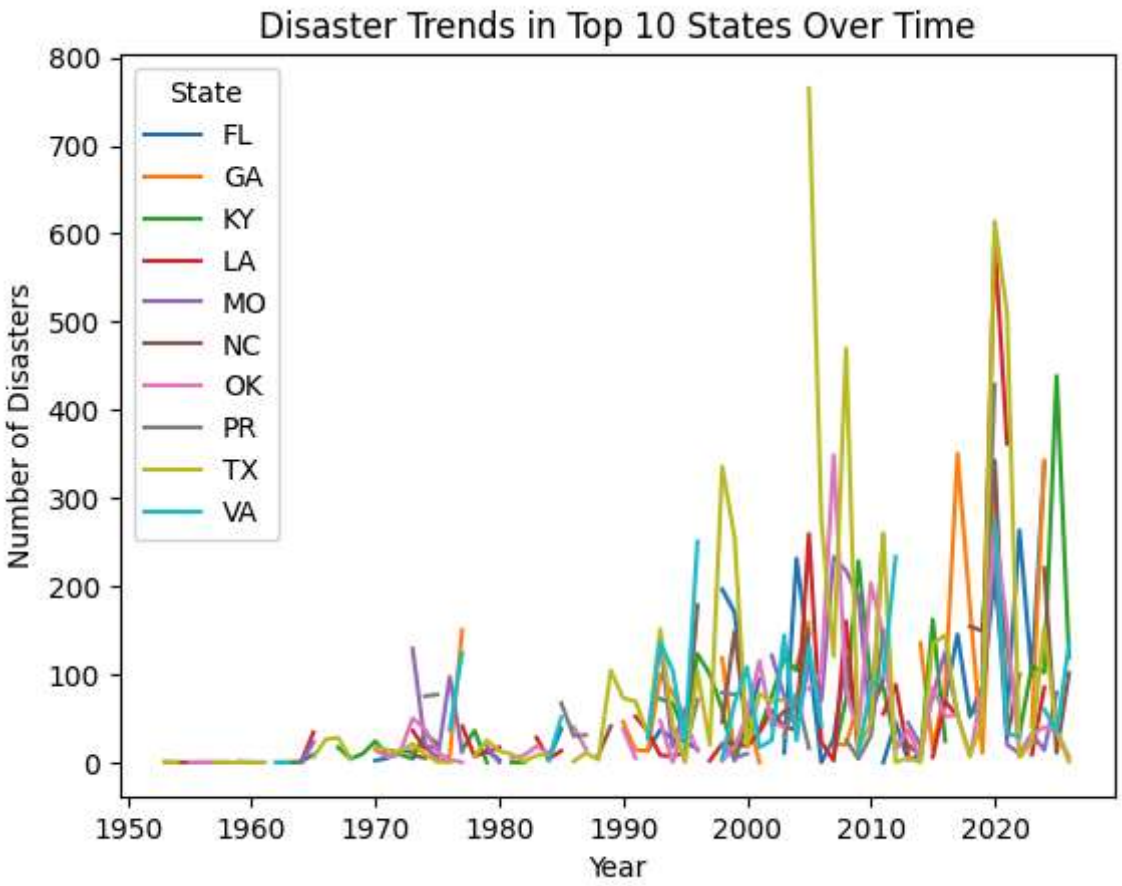
Step 3 — Group by Year and State

In [19]:
```
state_year_trend = top_states_df.groupby(['year','state']).size().unstack()
```

Step 4 — Plot Multi-Line Chart

In [20]:
```
plt.figure(figsize=(12,6))

state_year_trend.plot()

plt.title("Disaster Trends in Top 10 States Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Disasters")

plt.legend(title="State")

plt.show()
```

```
<Figure size 1200x600 with 0 Axes>
```



Observation: The chart shows disaster trends for the top 10 states with the highest number of disaster declarations.

Pattern: States like Texas, California, and Florida consistently experience higher disaster declarations compared to others.

Possible Reason: These states are more exposed to natural hazards such as hurricanes, wildfires, and floods.